EvOLve Forest Intelligence Report

Wayanad Wildlife Sanctuary (Muthanga Range) · 2019-07 to 2025-12

Generated: 13 August 2026, 22:00 IST

1. Executive Summary

The EvOLve system monitored 64 forest patches across the study area covering 6 years (2019–2025) using 72 months of Sentinel-2 satellite imagery. 7 patches (10.9%) showed evidence of forest degradation based on the self-supervised encoder + Genetic Algorithm adaptive thresholds.

2. Model Performance (Benchmark Comparison)

Model	Accuracy	F1 Score	AUC	Precision	Recall
EvOLve (self-supervised)	0.9688	0.8571	0.9925	0.8571	0.8571
Supervised Baseline	0.9688	0.8333	0.9975	1.0	0.7143
Random Forest	1.0	1.0	1.0	1.0	1.0

3. Wildlife Corridor Status

Of 8 north-south wildlife corridors analysed: 6 are Intact, 2 are Weakened, and 0 are Broken. Broken corridors restrict elephant and tiger movement — immediate intervention is recommended.

4. Fire Risk Assessment

Fire risk assessment (NDVI + SWIR + seasonal analysis): 0 patches Critical, 0 High, 12 Moderate, 52 Low risk. Pre-fire season monitoring (Feb–May) is critical for Wayanad.

5. Carbon Stock Estimation

Total carbon stock: 789,050.6 tons CO■ eq (est. USD 11,835,759). Annual carbon loss due to degradation: 17,788.5 tons CO■ (est. USD 266,828/year at USD 15.0/ton).

6. Encroachment Alerts

Total anomalies detected: 195 (25 High, 54 Medium, 116 Low severity). High severity events indicate sudden NDVI drops inconsistent with seasonal patterns — likely indicators of illegal clearing or rapid encroachment.

7. Reforestation Priorities

Rank	Patch ID	Priority Score	Degradation	Justification
1	41	0.8164	0.8937	Priority: high degradation (0.89); would restore wildlife co...
2	8	0.8065	0.9349	Priority: high degradation (0.93); would restore wildlife co...
3	16	0.7935	0.9289	Priority: high degradation (0.93); would restore wildlife co...